

NAME(S):

While Loops

Begin the following as paired-programming in class, but be sure to individually finish any problems you don't get to (hand in written answers of **0.25 engagement points**, check website for due date)

Learning goals / relevant topics

- Conditionals (Runestone Ch. 4)
- Comparison Operators (Runestone Ch. 2)
- While Loops (Runestone Ch. 6)

1 "Pencil/Pen and Paper" questions

Please try to solve these problems without running/writing code. Instead you should try to come up with your answers by tracing through the code yourself and reasoning about what the Python interpreter does. Note: You will see similar questions on the Pen and Paper Midterm.

1. What gets printed by the following code?

```
1) x = 10
2) while x > 8:
3)     print(x, end = " ")
4)     x = x - 1
```

.....

2. What gets printed by the following code?

```
1) x = 10:
2) while x > 8:
3)     x = x - 1
4)     print(x, end = " ")
```

.....

3. What gets printed by the following code?

```
1) i = 0
2) while i < 3:
3)     print(i, end = " ")
```

.....

4. What gets printed by the following code?

```
1) valid = False
2) while not valid:
3)     x = int(input("Enter a number: "))
4)     valid = (x % 2 == 1 and x % 3 == 0)
```

.....
5. How can we translate this code into a while loop?

```
1) for i in range(n):
2)     <body>
```

.....
.....
.....

Reminders for Pair Programming

- Instructions:
 - driver (has the keyboard and types) and navigator (guides the driver)
 - You should be talking to each other constantly!
 - You should be able to swap roles at any moment.
 - Swap roles every 5 to 10 minutes.

2 Summing up numbers

Write a function called `cash_register` that repeatedly asks the user to input (positive) numbers and sums them all up. When the user inputs a negative number, the function should print out a message to let the user know their total. The negative number should not be included in the sum.

For example, one run of the function might look like this:

```
>>> cash_register()
Welcome to the cash register!
Please enter the price of each of your items.
Please enter a negative number to indicate when you're finished.

Please enter the price. (Input a negative to stop.) 2
Please enter the price. (Input a negative to stop.) 3.75
Please enter the price. (Input a negative to stop.) 9
Please enter the price. (Input a negative to stop.) 6
Please enter the price. (Input a negative to stop.) -2
Your total is: 20.75
```

3 Reeborg: Around the world

Use the world “Around - variable” from the “Introduction” collection of worlds. Every time the world is reloaded, it gives you a rectangular world of a different size. Additionally, Reeborg is carrying an infinite number of tokens. Write a program that makes Reeborg walk around the perimeters of the world exactly once, no matter how big the world is.

Hint: is there a way to mark the square(s) that you’ve been to before?

4 Reeborg: mountain climbing

Download the `mounatin_top` folder of starter code from the course website. It should include four different worlds, each of which starts with Reeborg at the foot of a mountain and the goal is to help Reeborg climb to the top (Note: Reeborg should climb not fly). Write a program using function definitions, loops, and conditional statements that help Reeborg climb the mountain in any of the four worlds that are given and in any other similar mountain environments.

All programs should include appropriate comments and should be organized in a reasonable fashion.